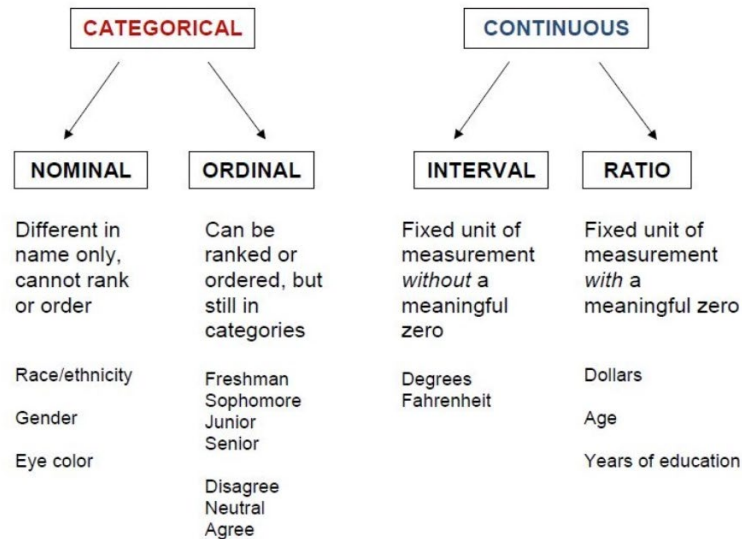


Changing Measurement Levels in JMP

As you undoubtedly know, there are four classical levels of measurement for all data: nominal, ordinal, interval, and ratio. See the graphic below.¹

In our class we typically have variables that are either required to be nominal or are just assumed to be continuous. We sometimes have to change the measurement level because even if it is set to be the right level on the .xlsx file, it has been our experience that



sometimes JMP will change that level once the data are imported. Regardless, it is a good skill to have to know to check and to know how to change the level in the software you are using.

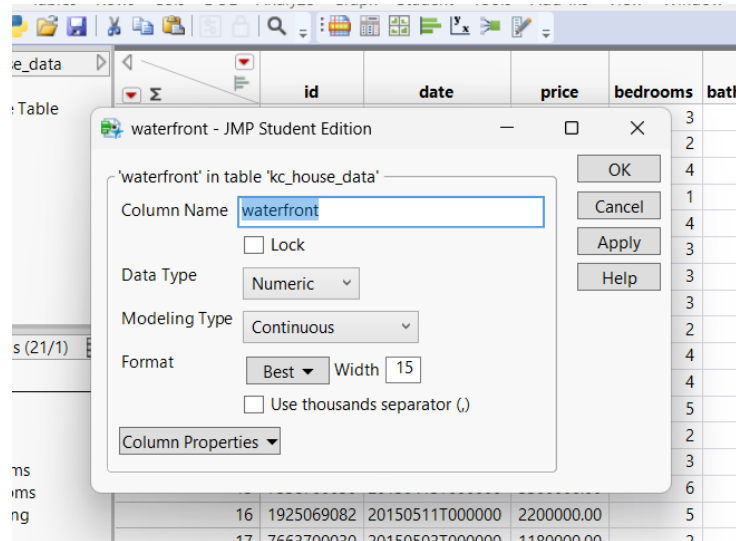
Let's look at an example. Below is a small snippet from the so-called Kansas City House Price Data after being loaded into JMP.

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view
1	822039084	20150311T000000	1350000.00	3	2.5	2753	65005	1	1	2
2	809600060	20150413T000000	655000.00	2	1.75	1450	15798	2	1	4
3	2025069065	20140929T000000	2400000.00	4	2.5	3650	8354	1	1	4
4	2123039032	20141027T000000	369900.00	1	0.75	760	10079	1	1	4
5	3225069065	20140624T000000	3080000.00	4	5	4550	18641	1	1	4
6	2122039094	20141126T000000	705000.00	3	3	1970	20978	2	1	3
7	622049114	20150218T000000	2120000.00	3	2.5	5403	24069	2	1	4
8	3760500116	20141120T000000	3070000.00	3	2.5	3930	55867	1	1	4
9	7567600045	20140827T000000	825000.00	2	1	1150	12775	1	1	4
10	4055701200	20150421T000000	1960000.00	4	2.75	3120	7898	1	1	4
11	8069000075	20141229T000000	790000.00	4	1.75	2460	10061	1	1	4
12	1247600105	20141020T000000	5110000.00	5	5.25	8010	45517	2	1	4
13	3523029041	20141009T000000	290000.00	2	0.75	440	8313	1	1	3
14	7638000041	20140625T000000	995000.00	3	4.5	4380	47044	2	1	3
15	7558700030	20150413T000000	5300000.00	6	6	7390	24829	2	1	4
16	1925069082	20150511T000000	2200000.00	5	4.25	4640	22703	2	1	4
17	7663700030	20150503T000000	1180000.00	2	2.5	1770	7155	2	1	4
18	8073000550	20150415T000000	1700000.00	4	3.75	3190	17186	2	1	4
19	8888000055	20141230T000000	530000.00	3	0.75	920	20412	1	1	2
20	1925069121	20150330T000000	960000.00	3	2.5	1730	4102	3	1	4
21	1922039062	20150420T000000	480000.00	2	1.5	1008	26487	1	1	4
22	3623500260	20140512T000000	1200000.00	3	1.75	1560	8078	1.5	1	4
23	8106100105	20141114T000000	3850000.00	4	4.25	5770	21300	2	1	4
24	1224059053	20141027T000000	1700000.00	5	2	2500	15250	2	1	4
25	3343301920	20150302T000000	1650000.00	3	2.75	2690	8890	2	1	4
26	1724069062	20140714T000000	1360000.00	2	3.25	2700	3444	3	1	3
27	7524900003	20141210T000000	3280000.00	2	1.75	6840	10000	2.5	1	4
28	7738500731	20140815T000000	4500000.00	5	5.5	6640	40014	2	1	4
29	4114601570	20141118T000000	3600000.00	3	3.25	5020	12431	2	1	4

¹ With thanks to National University's Academic Success Center: <https://resources.nu.edu/statsresources/measurement>

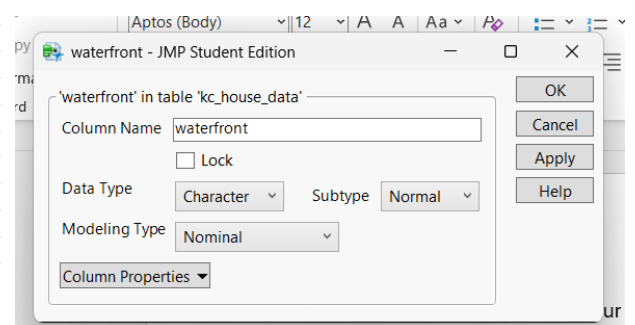
The variable “Waterfront” is there to denote whether the property record refers to a lot that has a waterfront. That’s either a yes or a no. In the dataset though, the variable is being treated as continuous. There are two clues to that in JMP. The first is that the data column is right justified. It will be left justified if it is nominal. The second clue is found on the lower left under the Columns. Nominal variables will show up in Red, continuous in Blue.

To change this, right click on the Waterfront column heading, and click Column Information. You’ll see this:



Change Data Type to Character and Modeling Type to Nominal. The latter may happen automatically.

Click OK



Notice the change (in justification and color).

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront
1	822039084	20150311T000000	1350000.00	3	2.5	2753	65005	1	1
2	8096000060	20150413T000000	655000.00	2	1.75	1450	15798	2	1
3	2025069065	20140929T000000	2400000.00	4	2.5	3650	8354	1	1
4	2123039032	20141027T000000	369900.00	1	0.75	760	10079	1	1
5	3225069065	20140624T000000	308000.00	4	5	4550	18641	1	1
6	2122039094	20141126T000000	705000.00	3	3	1970	20978	2	1
7	622049114	20150218T000000	212000.00	3	2.5	5403	24069	2	1
8	3760500116	20141120T000000	307000.00	3	2.5	3930	55867	1	1
9	7567600045	20140827T000000	825000.00	2	1	1150	12775	1	1
10	4055701200	20150421T000000	196000.00	4	2.75	3120	7898	1	1
11	8069000075	20141229T000000	790000.00	4	1.75	2460	10061	1	1
12	1247600105	20141020T000000	511000.00	5	5.25	8010	45517	2	1
13	3523029041	20141009T000000	290000.00	2	0.75	440	8313	1	1
14	7636800041	20140625T000000	995000.00	3	4.5	4380	47044	2	1
15	7558700030	20150413T000000	530000.00	6	6	7390	24829	2	1
16	1925069082	20150511T000000	220000.00	5	4.25	4640	22703	2	1
17	7663700030	20150503T000000	118000.00	2	2.5	1770	7155	2	1
18	8073000550	20150415T000000	170000.00	4	3.75	3190	17186	2	1
19	8888000055	20141230T000000	53000.00	3	0.75	920	20412	1	1
20	1925069121	20150330T000000	96000.00	3	2.5	1730	4102	3	1
21	1822039062	20150420T000000	48000.00	2	1.5	1008	25407	1	1